8*10/100/1000Base-T, 8*100/1000Base-X SFP

4*1G/2.5G/10GBase-X SFP+

L3 Managed Industrial Ethernet Switch with DI/DO/RS485

LT-IS6520XG-4XS8GX8GT-D

Features:

- ➢ 8*10/100/1000Base-T RJ45, 8*100/1000Base-X SFP, 4*1G/2.5G/10GBase-X SFP+
- ➢ Redundant power 12-75V input with polarity reverse/over-current protection
- ➢ Support DI/DO/RS485 and USB one key configuration & upgrading
- ➢ Layer 3 features: Support IPv4/IPv6 static routing, DHCP server, DHCP relay; Support ECMP, VRRP, RIP, OSPF, BGP4, IS-IS; 13K routing entries
- ➢ Layer 2 features: VLAN/GVRP/QinQ/SPAN/RSPAN/STP, RSTP, MSTP/DHCP/Multicast/ACL QoS/LLDP/802.1X/AAA/Loop Detect/Dying Gasp/SFP DDM/IPV6 management/Dual Stack/ sFlow/Web/SNMP/MQTT/Telnet/TFTP/Web upgrading
- ➢ Support G.8032 ERPS protocol, recovery time ≤20ms
- ➢ Support 6KV surge protection and ESD: Air-15kV, Contact-8kV Protection
- ➢ IP40 fan-less and Din-rail hardware design
- ➢ Operation temperature: -40 ℃ ~+75℃

Overview

The SmartByte LT-IS6520XG-4XS8GX8GT-D is the highly reliable managed industrial Ethernet switch with 8-port 10/100/1000-T, 8-port 100/1000Base-X SFP, 4-port 1G/2.5G/10GBase-X SFP+. In normal operation, it is a L3 managed Ethernet switch. It supports ERPS redundant network and the self-recovery mechanism less than 20ms on full load which allows you to scheme a reliable Ethernet network by building a redundant ring topology as your back-up solution. It also supports DI/DO/RS485, DHCP server, DHCP relay, IPv4/IPv6 static routing and dynamic routing protocols like RIP, OSPFv2, BGP4, IS-IS, robust Layer 2 management and secure functions, which can help users expand their networks quickly and efficiently.

LT-IS6520XG-4XS8GX8GT-D is the high cost-effective easy-to-use device, which provides essential industrial Ethernet networking function, including wide range power input 12-75VDC, redundant power design with polarity reverse protection, robust IP40 fan-less housing with Din-rail installation, wide operation temperature from -40°C to 75°C as well as high-level EMI/EMC capability. It is the best choice for heavy industrial factory, transportation, oil & gas, chemical, IP Surveillance and processing automation area where environmental conditions are harsh and critical.

Technical specification

Model No.	LT-IS6520XG-4XS8GX8GT-D	
Interface	Fiber ports	Copper RJ45 ports
	12	8
Ethernet	8*10/100/1000Base-T RJ45 8*100/1000Base-X SFP (9-16 ports) 4*1G/2.5G/10GBase-X SFP+ (17-20 ports)	
CPU	Marvell AC5 series 1.2GHz	
Size of RAM	1GB	
Size of Flash	512MB	
Switch chip model	Marvell AC5 series	
Management port	1*RJ45 Console port (115200,8,N,1)， 1*USB 2.0	
Data & Alarm channel（Removable 8-pin terminal block)	Vout: VO + , VO -	5V output（0.2A MAX）
	1 channel RS485: A, B	**Serial server:** Supports Modbus protocol transparent transmission **Network Side:** Supports TCP protocol, supports client and server mode **Serial Port Side:** Supports RTU

		communication mode and serial port transmission parameter settings
	1 channel digital input: DI+，DI-	Status “1”: 12V-30V Status”0”: -30V~3V Max input current: 8 mA
	1 channel digital output	Normally open， Normally closed， Pulse 2A@30V DC，0.3A@125V AC
Standard	IEEE 802.3 10Base-T Ethernet IEEE 802.3u 100Base-TX Ethernet IEEE 802.3ab 1000Base-T Ethernet IEEE 802.3z 1000Base-X Ethernet IEEE 802.3ae 10Gigabit Ethernet IEEE802.3x Flow Control and Back Pressure IEEE 802.1D Spanning Tree Protocol IEEE 802.1w Rapid Spanning Tree Protocol IEEE 802.1Q VLAN ITU-T G.8032 ERPS IEEE 802.1X Port Authentication Network Control IEEE 802.1ab LLDP IEEE 802.3ad LACP	
LED Indicators		
P1/P2 (Power indicator)	Off: the device is power off or failed	On: the device power on is normal
SYS (System status indicator)	Blinking: device initialization	On: device on normal operation
USB (USB status indicator) Green	Off: USB flash drive not insert or not detected by device	
	On: USB flash drive is detected by device	
	Blinking: the device is reading and writing the data from USB flash drive	
ALM (Fault status indicator)	Off: Device on normal working state	On: Fault alarm
Ring (Ring network indicator)	Off: Ring network disabled	On: Ring network enabled
1-8 (Copper ports indicators) Green	Off: ports link down	
	On: ports link up	
	Blinking: data on TX/RX	
1-8 (Ports speed indicators) Yellow	Off: ports speed on 10/100M	
	On: port speed on 1000M	
9-20 (Fiber ports indicators) Green	Off: ports link down	
	On: ports link up	
	Blinking: data on TX/RX	

Power parameter	
Input voltage	12-75VDC redundant power input
Input current	1.8A Max
Power consumption	Full loading ≤22W
Connector	Removable 2*2 pin terminal block
Reverse polarity protection	Support
Over-current protection	Support
Layer 2 switching	
Port aggregation	Support GE port aggregation
	Support 2.5GE aggregation
	Support 10GE aggregation
	Support static aggregation
	Support LACP dynamic aggregation
	Up to 64 aggregation groups and up to 8 ports per group
Port features	Support IEEE802.3x flow control
	Support interface counters
	Support port isolation
	Port mirroring (One-to-One, Many-to-One)
	Support loop detection (Port-based; VLAN-based)
	Support broadcast storm suppression (broadcast; unknown multicast; unknown unicast)
VLAN	IEEE802.1Q tagged VLAN
	Mac based VLAN
	IP based VLAN
	Protocol based VLAN
	Voice VLAN, specially divide VLAN for user voice data flow
	Private VLAN, divide a VLAN broadcast domain into multiple subdomains to form a double-layer VLAN structure
	VLAN range: 1-4094
	Support access mode/trunk mode/hybrid mode
Mac-address table	Support static Mac-address management

management	Support dynamic Mac-address Management
	Support filtering Mac-address
	Support MAC limit based on port and VLAN
	Support MAC flapping based on port and VLAN
	16K Mac-address table size
GVRP	Normal mode
	Fixed mode
	Forbidden mode
QinQ	Port-based QinQ
	VLAN-based QinQ
	VLAN mapping
	Flow-based QinQ
LLDP	Link Layer Discovery Protocol
	Med
SPAN	Number of Sessions: 7
	Many-to-one
RSPAN	Remote Switch Port Analyzer
MSTP	802.1D STP
	802.1W RSTP
	802.1S MSTP
	BPDU Guard
	Number of Single Ring Device Connections: 20
ERPS G.8032	Major ring
	Sub ring
	Recovery time ≤20ms
	Number of Single Ring Device Connections: No limit
QoS	QoS Class, Remarking
	Support SP, WRR Queue Scheduling
	Ingress Port-based Rate-limit
	Egress Port-based Rate-limit

	Egress port-queue shaper
	Policy-based QoS
	Port Queue Numbers: 8
DHCPv4 Snooping	DHCP snooping trust port
	DHCP snooping option-82
	DHCP snooping circuit-id and remote-id customized
L2 Multicast	
IGMP	IGMP v1, v2, v3
IGMP Snooping	Internet Group Management Protocol Snooping
	Fast Leave
	Multicast Table Size: 1K
PIM-SM	Protocol Independent Multicast-Sparse Mode
Security features	
ACL	IP Standard ACL IP Extended ACL MAC extend ACL IPv6 ACL Time Range ACL apply on VLAN ACL entries, include input and output
802.1x	Port access control Mac-address access control RADIUS server
Port-security	Port-security
IP source guard	IP Source Guard, Port-based IP/Mac Binding Access
ARP-check	Support ARP-check and ARP packet filtering for illegal users
DAI	Dynamic ARP Inspection
Anti-gateway ARP Spoofing	Prevent Gateway ARP Spoofing
Ddos	Ddos attack protection
Access control	Support Telnet/SSH/HTTP/HTTPS user access control
AAA	RADIUS authentication access switch
	TACACS authentication access switch
	Authorization

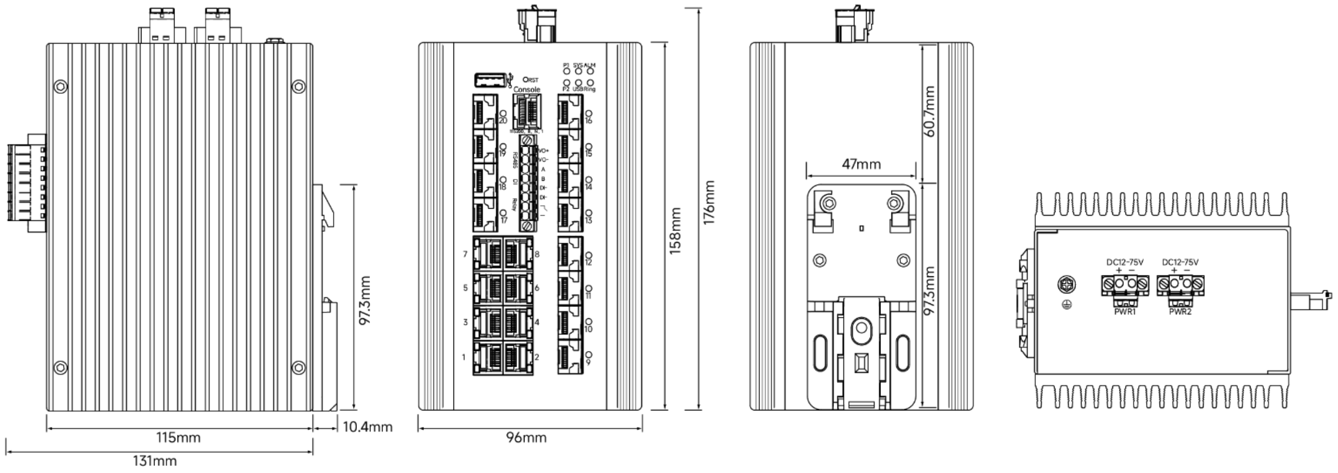
Reliability	
Loop Detect	Loop detect, based on port or VLAN
EFM	The Ethernet in the first mile, 802.3ah
Management & Maintenance functions	
Dual Stack	IPv4 and IPv6 Dual Stack Management
System Log Management	Syslog Level 7 to 0 Corresponding Debug/Info/Notice/Warning/ Err/ Critical/ Alert
	Support connecting up to 3 Servers
	For IPv4 and IPv6
User Management	Support password protection, support MD5/SHA256/SHA512 algorithm
	Local Management of User Privilege
	Remote authorization by RADIUS or TARCAS
	Enable privilege protection
	User password complexity requirements
	User login fail protection
	User Privilege Level 0 to 15
Login Service Management	Up to 8 users online simultaneously
	Access control based on IPv4 or IPv6 ACL
Firmware Management	Firmware upgraded by WEB
	Firmware upgraded by CLI
	Dual partition
Web Management	Web Management
	Support HTTP V1.1
	Support HTTPS
	For IPv4 and IPv6
CLI management	Console/Telnet command line management
Fault Detection	Ping/Traceroute, For IPv4 and IPv6
	Dying gasp (the input voltage should be more than 18V)
	Optical transceiver DDM
	One click fault information collection

	Cable detection
DNS	Domain Name System
NTP	Network Time Protocol, For IPv4 and IPv6
	Support domain name configuration
Configuration Import /Export	FTP/TFTP based remote configuration import and export
	USB flash drives based local configuration import and export
sFlow	Sampled flow
SNMP	Version SNMPV2C/V3
MQTT	Message Queuing Telemetry Transport
RMON	Support RMON (Remote Monitoring) alarm
Dual partition management	Support dual partition switching
Application Protocol Functions	
Telnet server	Support telnet server for IPv4 and IPv6
Telnet client	Support telnet client for IPv4 and IPv6
SCP (Secure Copy Protocol)	SCP client/server for IPv4 and IPv6
SFTP	sftp client/server for IPv4 and IPv6
SSH server	Support login based on user password-SSH2, support IPv4, IPv6
	Support public key-based login
TACACS	Support TACACS (the terminal access controller accesses the control system)
TFTP	Support TFTP Client
Layer 3 features	
Static Route	Support VLAN interface and route port
	IPv4 static routing table entry configuration
	ARP dynamic learning and static configuration
	IPv6 static routing table entry configuration
	Neighbor discovery and static configuration
	Routing prefix table size 4k for IPv4, for IPv6 divide by 4
DHCPv4 Client	Dynamic host configuration protocol for IPv4
DHCPv6 Client	Dynamic host configuration protocol for IPv6
DHCPv4 Relay	DHCP Relay for IPv4

DHCPv4 Server	DHCP Server for IPv4
ECMP	Support ECMP (Equal-cost multi-path routing)
	Support the configuration of ECMP Max next-hop
	Support capacity balanced configuration
Route policy	IPv4 prefix-list
VRRP	Virtual Router Redundancy Protocol
Routing Protocol Specification	
RIP	RIPv1/v2
OSPFv2	OSPFv2
BGP4	BGP4
	BGP Support Routing Recursive ECMP
	Support to view the number of neighbors and up/down state
IS-IS	IS-ISv4
Routing entries	13K
Switching features	
Switching capacity	112Gbps
Packet forwarding rate	166.6Mpps
MAC address table	16K
VLAN	4K
Buffer	12Mbit
Forwarding delay	<5us
Jumbo Frame	Support 10Kbytes
MDI/MDIX	Support
Watchdog	Support
Mechanical structure	
Case protection	IP40
Installation method	Din-rail
Dimension(W*D*H)mm	96*115*158mm
Weight	1.8 kg
Operating environment	

Operating temperature	-40℃~+75℃
Storage/transportation temperature	-40℃~+85℃
Relative humidity	Operation humidity: 10%-90%RH Storage humidity: 5%-95%RH
Industrial Standard	
EMS	Surge protection of power: IEC 61000-4-5 6KV/4KV（8/20us）
	Surge protection of Ethernet ports: IEC 61000-4-5 6KV/2KV（10/700us）
	RS：IEC 61000-4-3 80 MHz-1 GHz：10 V/m
	EFT：IEC 61000-4-4 4K/2K
	CS：IEC 61000-4-6 10V
	ESD：IEC 61000-4-2 Contact: 8K；Air: 15K
EMI	FCC Part 15B Class A
Shock	IEC 60068-2-27
Free fall	IEC 60068-2-32
Vibration	IEC 60068-2-6
Certification	CCC/CE/UKCA/RoHS/REACH/FCC
MTBF	> 200,000 hours
Warranty	5 years

Structure diagram


115mm
10.4mm
131mm
97.3mm
96mm
158mm
176mm
RST
Console
47mm
60.7mm
97.3mm
DC12-75V
DC12-75V
PWR1
PWR2

Order information

Model	Description
LT-IS6520XG-4XS8GX8GT-D	L3 Managed industrial Ethernet switch with 8*10/100/1000Base-T RJ45 ports , 8*100/1000Base-X SFP and 4*1G/2.5G/10GBase-X SFP+ slots, supports DI/DO/RS485, DC12-75V input, redundant dual power supply; Din-rail installation; Operation temperature: -40℃～+75℃